Analysis of Insurance Pricing Pre- and Post-Underwriting, Q1-Q3 2025

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Introduction

The purpose of this analysis is to examine how accurately the firm estimates pricing prior to full medical underwriting. The pre-underwriting pricing is based on a few basic components such as the insured's age, birth gender, assumed risk profile (Preferred, Standard, or various classes of Sub-Standard), and state of issue. The medical underwriting process involves a review of sensitive medical data as well as a detailed insurance physical. This can feel like an invasive process for many clients and there is often sensitivity around the final pricing offers they receive in response to the insurance market's assessment of their mortality risk.

A vital component of the firm's business model involves handling client expectations throughout the underwriting and implementation process. The life insurance industry is unique in that the firm maintains a relationship with clients for 40+ years, at a minimum, and often over multiple generations. Thus, it is vital to establish trust and rapport early in the relationship and to maintain that trust. A key part of establishing trust involves setting expectations that are grounded in reality. Failing to do so can cause frustration and can ultimately result in damage to the firm's relationship with the client.

This analysis seeks to determine if the firm is effective overall in setting pricing expectations in the initial pre-underwriting estimates. Then, we will examine which features are most predictive of the observed difference between Expected Cost and Realized Cost, which we hereafter refer to as the Cost Spread. We utilize data from 69 policies with policy dates between January 1, 2025 and September 30, 2025.

Analysis #1: Overall Effectiveness

The first analysis will analyze the median of the Expected Cost versus the median of the Realized Cost, both measured in basis points (bps) at life expectancy (see Definitions in the Appendix for a detailed explanation of these measurements). We are less concerned if the median of Realized Cost is *lower* than the median of Expected Cost, because this indicates that the firm is consistently delivering lower pricing than expected, which is a good outcome. Thus, our alternative hypothesis will examine only whether the median of Realized Cost is systematically *higher* than median of Expected Cost.

H_0: The median difference (median of Realized Cost minus median of Expected Cost) is ≤ 0

H_a: The median difference is > 0

Analysis #2: Factors Influencing the Cost Spread

The second analysis will examine some of the client data that was collected pre-underwriting and determine whether any of these factors are significant predictors of the size of the Cost Spread. These variables include basic metrics such as age, gender, assumed risk profile, and state of issue, as well as additional metrics such as the policy funding type (cash only or 1035 exchange of proceeds from another insurance policy), the size of the portfolio (expressed as a ratio relative to what the firm considers a "standard" size), as well as specific referral source. These factors are described in more detail in the Appendix.

We will produce an odds ratio for each factor to examine the strength of the relationship between the factor and the Cost Spread.

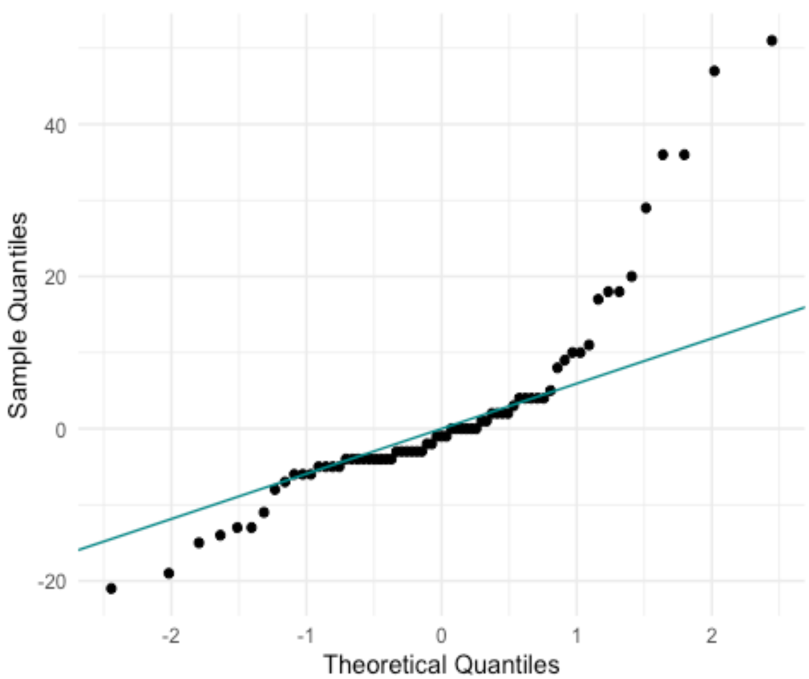
H_0: Each factor has *no effect* on the odds of the outcome of the Cost Spread (i.e. the odds ratio for each factor = 1)

H_a: Each factor has a *significant effect* on the odds of the outcome (i.e. the odds ratio for each factor $\neq 1$)

Methods

Shape of the Cost Spread Distribution

Fig. 1: Q-Q Plot of Pricing Dataset



The Cost Spread is *not* normally distributed. There are notable outliers on the higher end, which we consider an adverse outcome. A Shapiro-Wilk normality test confirms this result:

A Shapiro Wilks test confirms that the Cost Spread departs from normality ($W = 0.822$, p-value < 0.001).

Examination of High and Low Cost Spreads

Table 1: Comparison of Metrics for Observations with High Cost Spread vs. Low Cost Spread

Statistic	High Cost Spread	Low Cost Spread
Mean Age	50.3	46.2
Median Age	52.0	47.0
% Male	78.6	60.0
% Female	21.4	40.0
% Preferred Risk	50.0	69.1
% Standard Risk	50.0	30.9
% 1035 Exchanges	14.3	14.5
% Referred	50.0	21.8
Observations	14.0	55.0

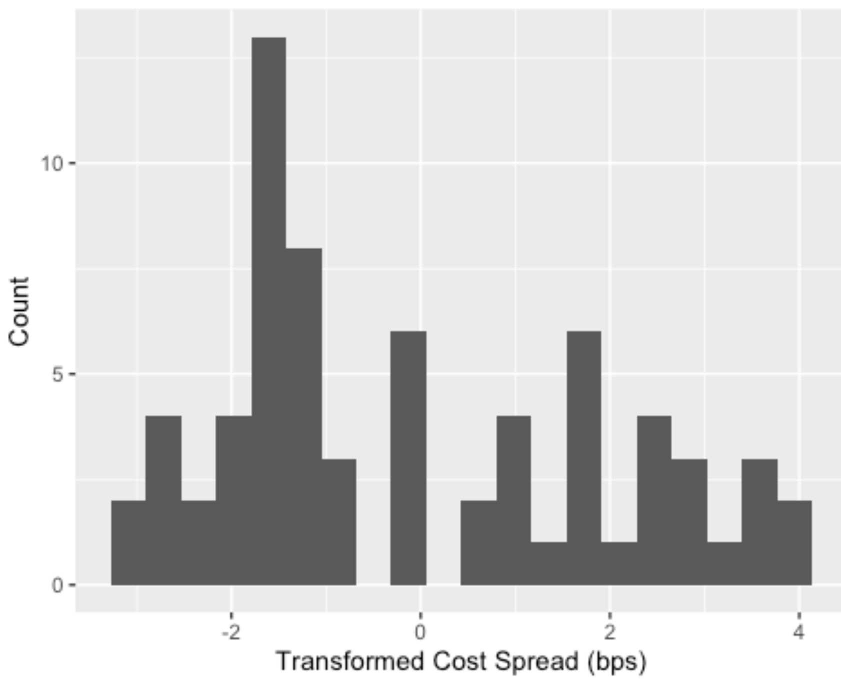
We define a "high cost spread" as observations where the Cost Spread is greater than 5 bps. There are clear patterns in cases with a high cost spread. Notably, these clients seem more likely to be older, male, assumed to be Standard risk pre-underwriting, and they tend to come from specific referral sources. However, the difference between the mean and the median for the high Cost Spread group seems to indicate that there is at least one client who is young but had a high Cost Spread. The gender difference may be due to the small size of dataset and the fact that clients are more likely to be male overall.

Distribution of the Transformed Cost Spread

Because we have paired, continuous data that is not normally distributed but is independent between pairs (the pricing for each client is determined by their own profile, uninfluenced by other clients), we would like to use the Wilcoxon signed-rank test to examine the overall effectiveness of the pricing estimates (Analysis #1).

To use this test, the distributions of Expected Cost and Realized Cost do not have to be normally distributed, but the distribution of the differences between them (the distribution of the Cost Spread) should be symmetric around the median. The Q-Q plot (Fig. 1) revealed a strong positive skew in the distribution of the Cost Spread. We will perform a signed log transformation to normalize the distribution while accounting for zero-values, as there are some observations for which the Realized Cost was equal to the Expected Cost (i.e. Cost Spread = 0).

Fig. 2: Histogram of Transformed Cost Spread



While this is not perfectly symmetrical around the median, it is notably more symmetrical than the non-transformed Cost Spread and we feel this is sufficient for the Wilcoxon signed-rank test.

Research Variables

Our analysis will utilize logistic regression to examine the relationship between the Cost Spread and various explanatory variables. Given the small sample size, converting the data to binaries seemed appropriate so that each subgroup has a meaningful number of observations for the regression.

Dependent Variable:

- Cost Spread: 0 for <= 5 bps, 1 for > 5 bps

Explanatory Variables:

- Age: Insured's age in years as of 1/1/2025
- Gender: 0 for male, 1 for female
- Assumed Risk Class: 0 for Preferred, 1 for Standard
- Issue State: 0 for standard issue states of AK, DE, SD; 1 for all other states
- Portfolio Size: 0 for portfolio size < 1, 1 for portfolio size >= 1
- Exchange: 0 for cash only, 1 for 1035 exchange

- Source: 0 for direct, 1 for referral

R Packages Used

The following packages were used in addition to base R: tidyverse (Wickham, 2023), ggplot (Wickham, 2016), lawstat (Gastwirth et al., 2024), knitr (Xie, 2015), odds.n.ends (Harris, 2021).

Results

Examination of Expected Cost vs. Realized Cost

Fig. 3: Distributions of Expected Cost and Realized Cost

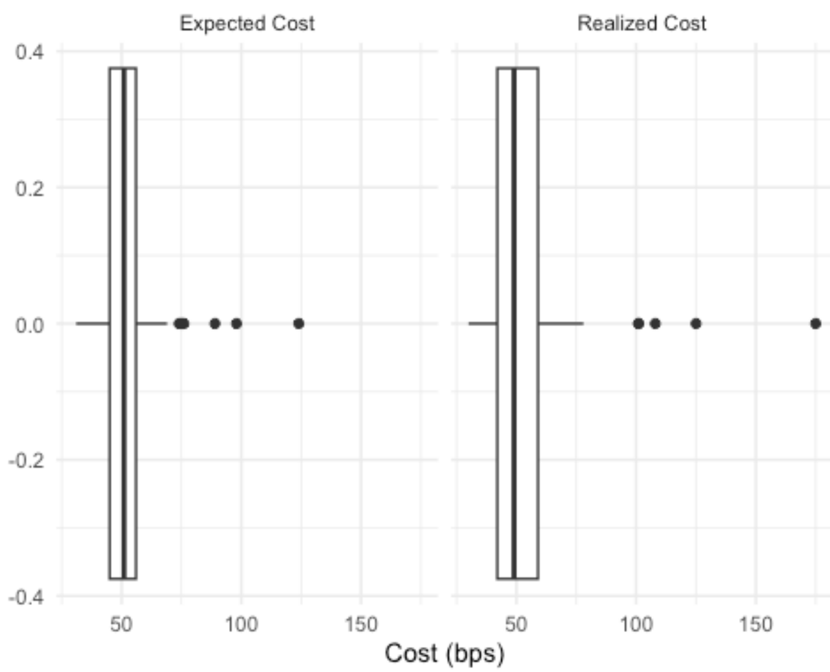


Table 2: Cost Spread Summary Statistics

Statistic	Expected Cost	Realized Cost
IQR	11	17
Minimum	31	30
25th Percentile	45	42
Median	51	49
75th Percentile	56	59
Maximum	124	175
Number of Observations	69	69

We note that the center of the distribution is relatively stable between Expected Cost and Realized Cost, even slightly lower, but the interquartile range is much wider. This is not unexpected given that only two risk classes were used to determine Estimated Cost (Preferred and Standard) and there are at least 18 potential Sub-Standard risk classes.

Analysis #1: Overall Effectiveness

A Wilcoxon signed-rank test confirms our observation regarding the stability of the median between Expected Cost and Realized Cost. With a p-value of 0.8963, we *fail to reject* the null hypothesis that the difference between the median of Expected Cost and the median of Realized Cost is less than or equal to zero. We conclude there is no evidence to suggest that Realized Costs are systematically higher than Expected Costs. On average, the firm appears to be accurately estimating costs during the pre-underwriting stage.

Analysis #2: Factors Influencing the Cost Spread

Fig. 4: Odds Ratios (95% Confidence)

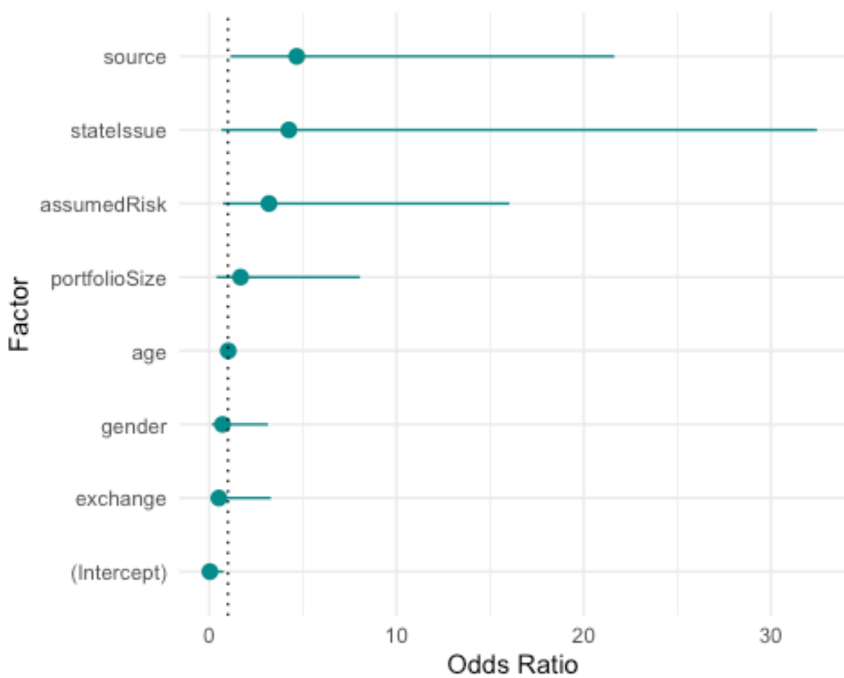


Table 3: Odds Ratios and 95% Confidence Intervals

Factor	Odds Ratio	Lower 95% CI	Upper 95% CI
(Intercept)	0.03	0.00	0.77
portfolioSize	1.67	0.38	8.05
age	1.01	0.95	1.08
gender	0.71	0.13	3.13
assumedRisk	3.19	0.73	16.03

Factor	Odds Ratio	Lower 95% CI	Upper 95% CI
stateIssue	4.25	0.65	32.45
exchange	0.52	0.05	3.30
source	4.67	1.14	21.64

Referral source appears to be the only significant factor in determining the Cost Spread, with a confidence interval that does not overlap 1, represented by the dotted line in Fig. 4. We can also see this in Table 3, which shows that both the lower bound and upper bound of our 95% confidence interval are greater than 1. State of issue and assumed risk are not significant but are borderline. Additional research should be conducted with a larger number of observations to determine if these factors might be significant.

Discussion

Research Outcomes

Our first analysis sought to determine if Realized Cost was systematically higher than Expected Cost. The Wilcoxon signed-rank test revealed no evidence to suggest that the median of Realized Cost was greater than the median of Expected Cost. We conclude that, on average, the firm is accurately estimating costs in the pre-underwriting stage.

The second analysis examined various factors that could potentially influence the Cost Spread. The logistic regression revealed no significant impact from portfolio size, age, gender, or fund source (cash only or 1035 exchange from another insurance policy). We conclude that there is no evidence that these factors influence the Cost Spread.

State of issue and assumed risk pre-underwriting are borderline factors; they are not statistically significant, but are very close. More data is needed to determine if these factors are truly significant or not.

The data suggests that referral source had significant impact on the Cost Spread. This factor is not related to medical underwriting directly, but rather product availability and pricing. Some referral sources have a small list of approved insurance companies, so the firm is unable to negotiate with the entire market to obtain best pricing. Additionally, these sources have custom pricing schedules that may create price dynamics that reflect a greater degree of change if the mortality risk is downgraded. Further investigation would need to be conducted to determine the exact factors making these referral sources more likely to impact the Cost Spread, and which referral sources have the most impact.

Broader Context

The size of the Cost Spread can have measurable financial impact on the firm. Pricing is a major obstacle during the sales process. One of the key aspects of managing the sales pipeline is to reduce ‘friction’ in the transaction process (Bertini, 2022). The more friction points there are in the sales process, the more opportunities there are for the firm to lose the business.

The pre-underwriting pricing review is an unavoidable point of friction, but it has the benefit of facilitating a "quick no". At this point, the firm has made no investment in the extensive (and expensive) medical underwriting process, which incurs costs that are borne by the firm alone and not directly billed to the client. The expectation is that these costs will be recouped with the commissions from the sale. If the client says no before underwriting, there is minimal financial impact to the firm. If the client says no *after* underwriting, the firm loses money on the investments made in underwriting that client.

If the post-underwriting cost is significantly higher than expected, then this introduces a new point of friction that increases the likelihood that the client does not move forward with the acquisition of a policy, with major financial cost to the firm. Thus, efforts to reduce the Cost Spread have a direct impact on the financial success of the firm.

Outstanding Questions

We have noted that state of issue and assumed risk pre-underwriting were borderline non-significant in our logistic regression. It may be the case that the small sample size of 69 observations did not provide enough data to determine whether these factors are truly significant or not. Additional research should be conducted with a larger number of observations so that we can properly assess these borderline cases.

A larger dataset may also allow for more nuanced investigation of the referral source factor and the elements within the referral source variable that may influence its impact on the Cost Spread. It would be worthwhile to parse out whether the product availability or the custom pricing schedule has a larger impact, or if some referral sources have a larger impact on the Cost Spread than others.

Additionally, this dataset only examined *inforce* policies and is thus only capturing clients who moved forward with obtaining a policy. It is possible that this has caused us to underestimate the Cost Spread. There may be cases where the Cost Spread was large enough that the client decided not to purchase a policy after completing medical underwriting. These cases are not included in our analysis. The firm tracks missed opportunities, and further research could expand the dataset to include this data along with the data related to policies placed inforce.

Conclusion

This paper sought to determine (1) whether the firm systematically underestimates pricing prior to medical underwriting and (2) which factors may be influential in determining the size of the post-underwriting Cost Spread. The data suggests that the firm is, on average, effective in estimating costs during the pre-underwriting stage of policy implementation. There is also evidence that referral source is a significant factor in determining the size of the eventual Cost Spread. Other factors, such as age, gender, and funding source did not appear to be significant based on our analysis. Assumed risk and state of issue are of borderline non-significance and additional data is necessary to parse out the potential significance of these factors. These analyses are important contributors to the financial success of the firm because they prevent the firm from losing business due to high Cost Spread after making investments in medical underwriting.

Appendix

Summary of the Dataset

The dataset consists of 69 insurance portfolios with policy dates between 1/1/2025 and 9/30/2025. Due to the proprietary nature of the data, much of the policy information, including product type and amount, has been removed from the dataset. A summary of the remaining anonymized variables is provided below.

- portfolioSize: Size of the client's portfolio in relation to what the firm considers a standard case, with size = 1 being equal to the standard case.
- age: Client's age as of 1/1/2025.
- gender: Client's gender at birth.
- assumedRisk: Risk class assumption in pre-underwriting pricing. Ranked as Preferred or Standard.
- stateIssue: State where the policy was issued, which often differs from the client's state of residence.
- stateResidence: Client's state of residence for underwriting and tax purposes.
- exchange: Whether the portfolio was funded, in whole or in part, by proceeds from an IRC Section 1035 tax-free exchange (Internal Revenue Service, 2025) of assets from one insurance contract to another (often between different insurance companies). 1 = Yes, the portfolio was funded in whole or in part by 1035 exchange proceeds. 0 = No, the portfolio was funded with new cash premium only.
- source: This represents specific referral sources that can have significant impact on pricing. Because these sources are confidential, they have been assigned a number to maintain anonymity.
- expectedCost: Cost illustrated in pre-underwriting analytics. This is defined as the difference, measured in basis points, between the assumed level annual rate of return, net of investment management fees, and the projected Internal Rate of Return (IRR) of the portfolio at life expectancy. Please see Definitions for more details about these measurements. If there are multiple policies in the portfolio, this is an aggregate value weighted the insurance benefit of each policy.
- realizedCost: Cost illustrated in the post-underwriting policy contracts. During the underwriting process, the insurance companies and their reinsurers review the client's medical records and provide pricing specific to the client's individual health

profile. The final pricing is reflective of the market's assessment of the client's risk profile as well as substantial negotiation between the firm, the insurance companies, and their reinsurers.

There was an additional column, 'smoker', which has been removed from the dataset. This column indicated whether the client was assumed to be a smoker in pre-underwriting pricing, with 1 = smoker and 0 = non-smoker. This variable did *not* indicate whether the client was a smoker, only whether this was disclosed to the firm prior to initiating medical underwriting. Of the 69 portfolios in the dataset, none were run with smoker pricing in pre-underwriting analytics.

Definitions

- Internal Rate of Return (IRR): The "discount rate that would make the net present value (NPV) of an investment zero." (Phalippou, 2024b) In simple terms, if I promised to pay you $100 next year, what discount rate would make the value of that payment $0 today? That discount rate is the rate of return. While IRR is not a perfect measure of performance (Phalippou, 2024a), it is a very common and easily understood measure of performance within the finance industry. It allows for comparison between investments with different time horizons and cash flows. The cash flows in this analysis are premium payments (outflow) vs. insurance benefit (inflow) received by the clients' heirs at their death (assumed to be life expectancy as defined below).

- Life Expectancy: expectedCost and realizedCost are measured by an inflow on the date of the insured's death, assumed in this analysis to be age 90. This assumption is based on proprietary research using the Society of Actuaries' 2008 VBT Select mortality tables (Kupstas, n.d.; **vba2008?**) as well as research from M Financial Group (M Financial Group, 2025) that adjusts the basic mortality tables to reflect lower mortality rates among the specific population represented by the clients in this dataset (Gong et al., 2016). Life expectancies, defined herein as the year in which the probability of the insured being alive is 50% based on the Society of Actuaries' 2008 VBT mortality table with adjustment, cluster around age 90 for most clients (within a range of 2-4 years). Therefore, for simplicity, comparability, and ease of understanding, this analysis assumes a life expectancy of age 90 for all insureds.

- Basis Point: Equal to 1/100th of 1%, or 0.01. Basis points are a common unit of measurement in financial analysis because they allow for more clarity in discussions around differences that, while less than 1%, can cause significant impact in outcomes over long time horizons or for large deposit amounts. (Corporate Finance Institute, 2023)

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